Connectome Brain Age: From Harmonization to Interpretation

DWI & fMRI

Ridge, GNN and multimodal fusion

ADNI · HABS · ADRC · ADDECODE (DWI only)

An evidence-based summary of existing analyses

Editable text, tables and native PowerPoint charts; source details in speaker notes.

Three findings frame the current results

- Harmonization substantially reduces dataset and site predictability; biological signal preservation is mixed.
- Ridge outperforms GNN in the reported unimodal analyses and the shared paired-sample comparison.
- Weighted Ridge fusion has the lowest paired-sample MAE: 6.47 years; its gain over DWI Ridge is modest.
- Available attribution is archived; the current code uses original saved OOF checkpoints, whose verified outputs are pending.

report_tables/Table4_models_pooled.csv; GNN_interpretability/analysis_summary.json

One pipeline, two evaluation populations

01 Harmonize

84 × 84 connectomes

3,486 unique edges

Joint dataset/site ComBat

02 Predict

CN normative age models

Ridge and GNN

Single modality + fusion

03 Interpret

BAG and age bias

OOF fusion GNN IG

ROI / edge / modality

Full modality cohorts answer a different question from the shared 387-scan fusion benchmark.

README.md; report_tables/Table1_samples.csv; Table2_features.csv

Data preparation and quality-control pipeline

1 Discover inputs

$A_i \in \mathbb{R}^{84x84}$

Connectome matrices + cohort metadata

2 Standardize IDs

key = dataset + participant + visit

Harmonize cohort-specific identifiers

3 Check matrices

shape(A_i) = (84, 84)

Require numeric, finite entries

4 Extract unique edges

$x_i = vec_u(A_i)$

$p = 84 \times 83 / 2 = 3{,}486$

Upper triangle; exclude the diagonal

5 Align covariates

$X \in \mathbb{R}^{nx3486}$

Match age / sex / diagnosis; retain audit

NOTATION AT A GLANCE

i; n Scan index; number of eligible scans

N; A_i N = 84 ROIs; A_i is 84 × 84

vec_u; x_i Upper triangle; $x_i \in \mathbb{R}^{3486}$

Output: auditable eligible scans, aligned features and DATASET_SITE batches.

Data preparation and quality-control: notation

Symbol / object	Definition and dimension	How to interpret it
i; n	Scan index; number of eligible scans	A participant may contribute multiple visits; i is not a participant index.
N; A_i	$N = 84$ ROIs; A_i is 84×84	A matrix entry is connectivity between two ROIs, not a regional volume.
vec_u; x_i	Upper triangle; $x_i \in \mathbb{R}^{3486}$	Take each unordered ROI pair once; omit the diagonal.
d; X	$d = N(N-1)/2 = 3{,}486$; X is $n \times d$	Rows are scans; columns refer to the same ROI pair across scans.
$q(i)$; visit(i)	Participant identifier; visit identifier	$q(i)$ defines grouping in cross-validation; visits remain together.
$b(i)$; c_i	Technical batch; biological design vector	$b(i)$ combines dataset and site; c_i encodes covariates to preserve.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Cohort sizes and the paired evaluation set

Analysis	Scans	Participants / scope
DWI harmonization	1,234	ADNI 314; HABS 487; ADRC 347; ADDECODE 86
fMRI harmonization	847	ADNI, HABS, ADRC
DWI CN OOF models	916	667 participants; HABS all-CN override
fMRI CN OOF models	642	525 participants
Paired DWI + fMRI	847	679 participants
Shared fusion CN OOF	387	330 participants

CN = cognitively normal; OOF = out-of-fold. Cohort definitions are policy dependent.

report_tables/Table1_samples.csv; DWI metadata; modality CV summaries

DWI harmonization pipeline

1 Structural edges

$x_i = vec_u(A_i)$

Four cohorts; 3,486 unique connections

2 Transform weights

$u_{ij} = \log(1 + x_{ij})$

Reduce skew; apply edge QC

3 Model batch effects

$u_{ij} = \alpha_j + c_i^T\beta_j + \gamma_{\beta j} + \delta_{\beta j}\,\varepsilon_{ij}$

b: dataset/site; c: age + sex in this saved run

4 Remove batch effects

$u^H_{ij} = (u_{ij} - \alpha_j - c_i^T\beta_j - \hat{\gamma}_{\beta j})/\hat{\delta}_{\beta j} + \alpha_j + c_i^T\beta_j$

Schematic ComBat; empirical Bayes estimates

5 Reconstruct matrices

$x^H_{ij} = \exp(u^H_{ij}) - 1$

Restore symmetric matrices; nonnegative export

NOTATION AT A GLANCE

x_{ij}; u_{ij} Raw weight; log1p weight, dimensionless

α_j; β_j; c_i Edge intercept; covariate coefficients; design vector

$\gamma_{\beta j}$; $\delta_{\beta j}$ Batch location shift; positive batch scale

Full-data harmonization is upstream of the current model CV; held-out age leakage remains possible.

DWI harmonization: notation

Symbol / object	Definition and dimension	How to interpret it
x_{ij}; u_{ij}	Raw weight; log1p weight, dimensionless	i indexes scans; j indexes unique connections; u = log(1+x).
α_j; β_j; c_i	Edge intercept; covariate coefficients; design vector	c_i contains age and sex in the saved DWI run; diagnosis is absent.
$\gamma_{\beta j}$; $\delta_{\beta j}$	Batch location shift; positive batch scale	b = b(i) is dataset/site, not the Ridge intercept.
σ_j; ε_{ij}	Edge residual SD; standardized residual	Feature standardization precedes empirical Bayes batch estimation.
$\gamma^*_{\beta j}$; $\delta^*_{\beta j}$	Empirical Bayes batch parameter estimates	Shrink noisy batch estimates using information across connections.
H; expm1	Harmonized quantity; inverse log1p	Reconstruct symmetric weights and apply the export nonnegative convention.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

fMRI harmonization pipeline

1 Functional edges

$r_i = vec_u(A_i)$

ADNI / HABS / ADRC; 3,486 correlations

2 Fisher-z transform

$z_{ij} = \text{arctanh}(r_{ij})$

Clip correlations before transformation

3 Model batch effects

$z_{ij} = \alpha_j + c_i^T\beta_j + \gamma_{\beta j} + \delta_{\beta j}\,\varepsilon_{ij}$

c: age, sex, diagnosis; b: dataset/site

4 Adjust in z-space

$z^H_{ij} = (z_{ij} - \alpha_j - c_i^T\beta_j - \hat{\gamma}_{\beta j})/\hat{\delta}_{\beta j} + \alpha_j + c_i^T\beta_j$

Schematic ComBat; preserve modeled biology

5 Restore correlations

$r^H_{ij} = \tanh(z^H_{ij})$

Rebuild symmetric matrices; diagonal = 1

NOTATION AT A GLANCE

r_{ij}; z_{ij} Correlation; Fisher-z feature

c_i; β_j Covariate design vector; edge coefficients

α_j; b(i) Feature intercept; dataset/site batch index

Output: 847 harmonized fMRI scans with traceable metadata.

fMRI harmonization: notation

Symbol / object	Definition and dimension	How to interpret it
r_{ij}; z_{ij}	Correlation; Fisher-z feature	r is clipped inside (−1,1); z = arctanh(r).
c_i; β_j	Covariate design vector; edge coefficients	Age is continuous; sex and diagnosis are categorical design columns.
α_j; b(i)	Feature intercept; dataset/site batch index	Each eligible scan belongs to one batch; singleton batches are removed.
$\gamma_{\beta j}$; $\delta_{\beta j}$	Batch shift; batch scale	Adjustment removes modeled technical location and scale effects.
σ_j; γ*; δ*	Residual SD; empirical Bayes batch estimates	The detailed ComBat diagram shows standardization and reconstruction.
z^H; r^H	Adjusted z feature; adjusted correlation	$r^H = \tanh(z^H)$; reconstruct an 84 × 84 symmetric matrix with diagonal 1.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

ComBat: standardize, adjust, restore

1 Remove modeled biological mean

$q_{ij} = \hat{\alpha}_j + c_i^T \hat{\beta}_j$

$s_{ij} = (v_{ij} - q_{ij})/\hat{\sigma}_j$

v: log1p DWI or Fisher-z fMRI; $\hat{\sigma}$: pooled residual SD

2 Estimate technical batch effects

$s_{ij} = \gamma_{\beta j} + \delta_{\beta j}\, \varepsilon_{ij}$

$(\gamma^*_{\beta j}, \delta^*_{\beta j})$ = EB estimates

Empirical Bayes shrinks location / scale estimates across edges

3 Adjust standardized values

$s^H_{ij} = (s_{ij} - \gamma^*_{\beta j})/\delta^*_{\beta j}$

δ* denotes the estimated SD scale; divide by sqrt(variance)

4 Restore feature units

$v^H_{ij} = \hat{\sigma}_j\, s^H_{ij} + q_{ij}$

Then expm1 for DWI or tanh for fMRI; reconstruct matrices

i: scan; j: edge; b: dataset/site batch; c: preserved design; H: harmonized; *: empirical Bayes estimate.

Current ComBat uses all eligible scans before CV; the displayed prediction estimates retain upstream leakage risk.

run_fMRI_harmonization.py; saved DWI combat_covariates.csv

Harmonization targets technical variability

- QC and metadata matching precede feature extraction; each matrix yields 3,486 upper-triangle edges.
- DWI: log1p transformation before ComBat; fMRI: Fisher-z transformation before ComBat.
- Technical batch: DATASET_SITE, combining cohort and acquisition site.
- Biological covariates: age and sex; fMRI also preserves diagnosis. Saved DWI run preserves age and sex only.
- Adjusted edges are reconstructed into symmetric 84 × 84 connectomes.

README.md; run_DWI_harmonization.py; run_fMRI_harmonization.py

Harmonization evaluation pipeline

1 Align diagnostic inputs

$X_{before} \leftrightarrow X_{after}$

Use matching eligible scans per target

2 Inspect PCA

$T = (X - \mu) V_2$

Separate PCA views; color dataset / site

3 Measure technical signal

$BA = (1/K) \Sigma_k recall_k$

Predict dataset and dataset/site batch

4 Assess biological signal

$MAE = (1/n) \Sigma_i |\hat{y}_i - y_i|$

Age MAE; sex / diagnosis balanced accuracy

5 Compare before / after

$R^2 = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$

Lower technical signal; assess biological loss

NOTATION AT A GLANCE

X; μ; V_2 Feature matrix; column means; two PCA loadings

K; $recall_k$ Number of classes; $TP_k/(TP_k+FN_k)$

y_i; $\hat{y}_i$; $\bar{y}$ Actual age; estimated age; evaluation mean

Assess reduced technical predictability together with retained biological information.

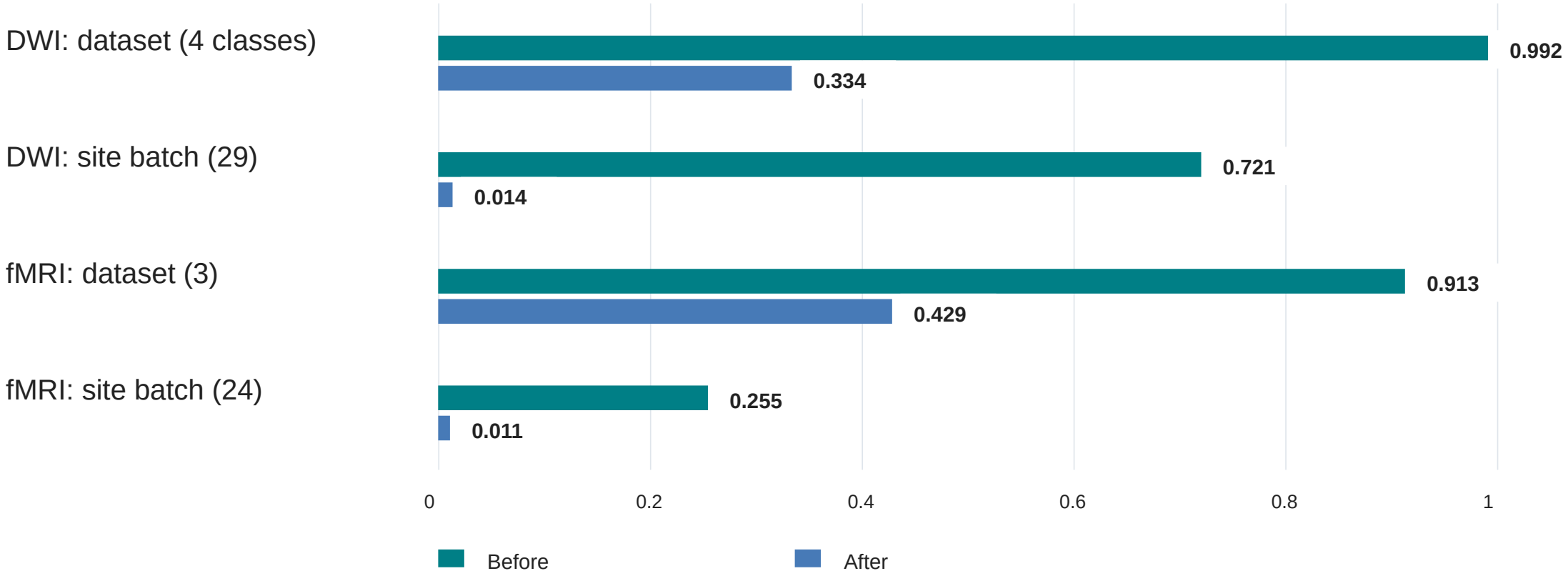
dwi_evaluation.py; fMRI_evaluation.py; report_tables/Table3_harmonization.csv

Harmonization evaluation: notation

Symbol / object	Definition and dimension	How to interpret it
X; μ; V_2	Feature matrix; column means; two PCA loadings	$T = (X-\mu)V_2$ is $n \times 2$; separate before/after fits need not share axes.
K; recall_k	Number of classes; $TP_k/(TP_k+FN_k)$	Balanced accuracy averages sensitivity equally across classes.
y_i; $\hat{y}_i$; $\bar{y}$	Actual age; estimated age; evaluation mean	Age and age errors are measured in years.
MAE; RMSE	Mean absolute error; root mean squared error	MAE = mean $\lvert\hat{y}-y\rvert$; RMSE = sqrt(mean $(\hat{y}-y)^2$).
R^2	1 − residual sum of squares / total sum of squares	$R^2 < 0$ means worse squared error than predicting the evaluation mean.
Target-specific n	Eligible scans for each diagnostic target	Sample sizes and class counts differ; do not interpret all rows as one set.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

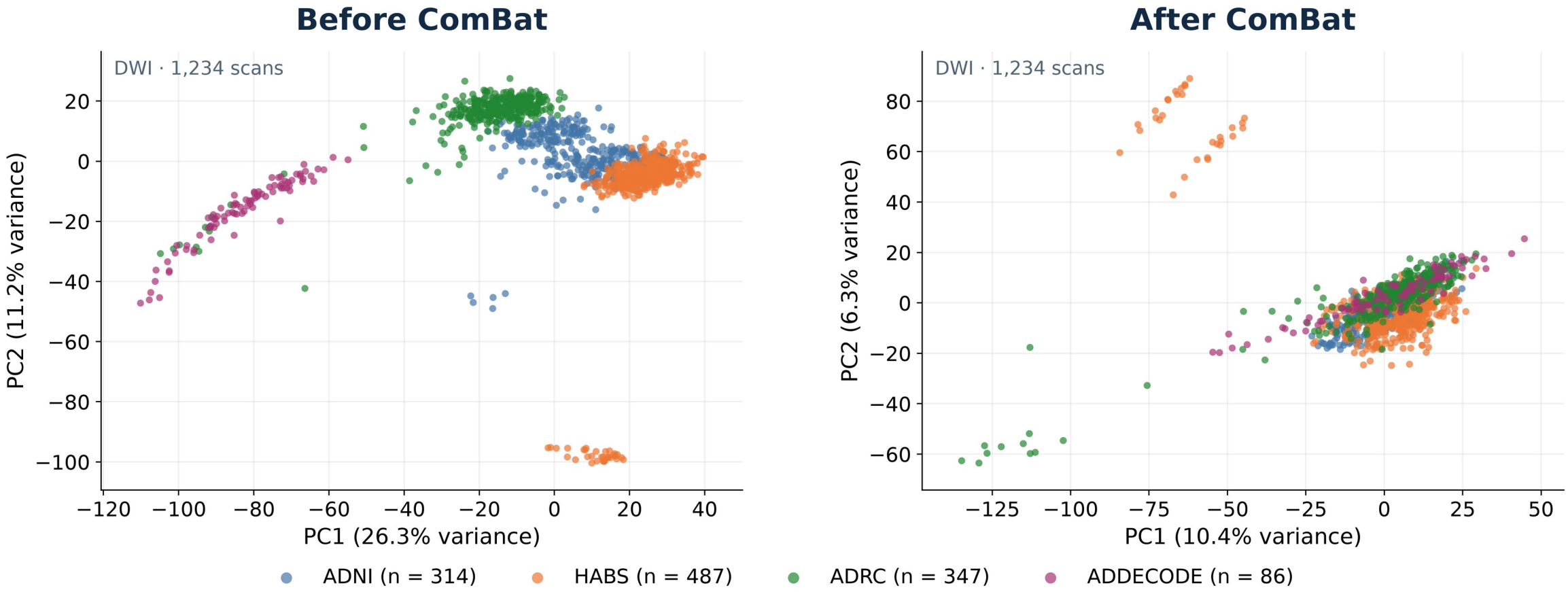
Dataset and site predictability decrease sharply



Balanced accuracy; lower technical predictability supports reduced batch information.

report_tables/Table3_harmonization.csv

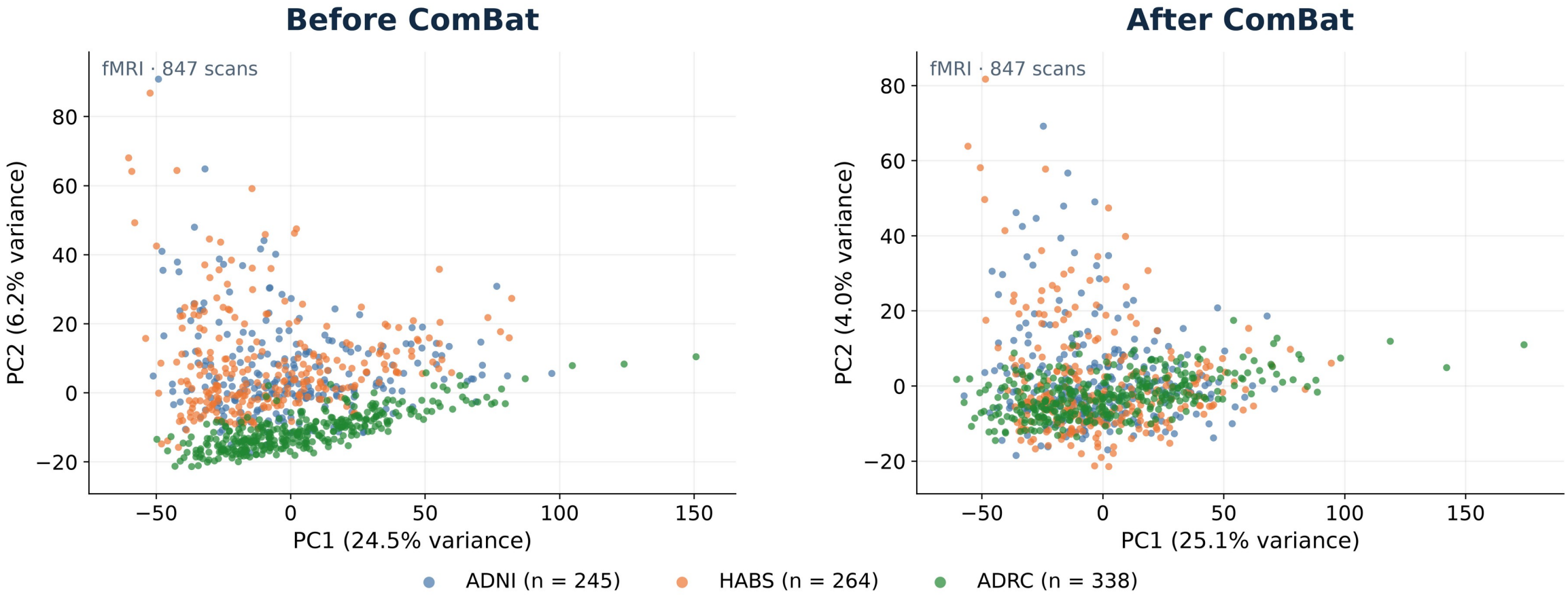
PCA by dataset: DWI



Separate standardized PCA fits; axes are specific to each panel. All eligible scans are shown.

results/checks/PCA/figures/DWI_PCA_by_dataset_before_after.png

PCA by dataset: fMRI



Separate standardized PCA fits; axes are specific to each panel. All eligible scans are shown.

Biological signal preservation is mixed

Diagnostic	DWI before → after	fMRI before → after
Sex balanced accuracy	0.782 → 0.766	0.631 → 0.640
Diagnosis balanced accuracy	0.294 → 0.151	0.300 → 0.227
Age MAE (years)	6.22 → 6.71	7.64 → 8.00
Age R^2	0.588 → 0.516	0.379 → 0.307

Sex prediction is broadly retained. Age prediction worsens and diagnosis prediction decreases; these diagnostics do not establish complete biological preservation.

report_tables/Table3_harmonization.csv

Evaluation uses participant grouping

- CN scans train normative age models; all visits from each participant stay within one outer fold.
- Ridge: five outer folds, four grouped inner folds for alpha selection; standardization is fitted within training splits.
- Unimodal GNN: graph sparsity is compared using pooled CN OOF performance.
- Paired benchmark: Ridge and GNN share 387 scans, 330 participants and the same outer folds.
- Upstream ComBat was fitted on all data while preserving age; current CV estimates retain leakage risk.

Ridge CV summaries; GNN threshold summaries; fusion_summary.json

Single-modality Ridge pipeline

1 Extract CN features

$x_i \in \mathbb{R}^{3486}$

DWI or fMRI; train on CN scans

2 Split by participant

train groups ∩ test groups = ∅

Five outer folds; visits stay together

3 Scale training features

$Z_{ij} = (x_{ij} - \mu_j)/\sigma_j$

μ / σ from each training split only

4 Tune Ridge penalty

min over β,b:
$\Sigma\text{train}(y_i - b - Z_i\beta)^2 + \lambda\|\beta\|_2^2$

Four grouped inner folds; b unpenalized

5 Predict held-out age

$\hat{y}_i = \hat{b} + Z_i\hat{\beta}$

Refit selected λ on outer-training data

NOTATION AT A GLANCE

x_i; Z_i Raw edge vector; standardized edge vector

μ_j; σ_j Training-column mean and standard deviation

$\beta \in \mathbb{R}^d$; $b \in \mathbb{R}$ Edge coefficients; unpenalized age intercept

All visits from a participant stay together; upstream ComBat remains outside this CV.

Single-modality Ridge: notation

Symbol / object	Definition and dimension	How to interpret it
x_i; Z_i	Raw edge vector; standardized edge vector	Both have d = 3,486 entries; DWI and fMRI are fitted separately.
μ_j; σ_j	Training-column mean and standard deviation	Re-estimate inside every inner split; outer-test observations never fit them.
$\beta \in \mathbb{R}^d$; $b \in \mathbb{R}$	Edge coefficients; unpenalized age intercept	$\hat{y} = b + Z\beta$; β converts standardized feature values into age years.
λ; $\lVert\beta\rVert_2^2$	Nonnegative penalty; sum of squared coefficients	λ is sklearn Ridge alpha; larger values shrink coefficients more strongly.
T_o; V_o	Outer-training set; outer-test set for fold o	All visits from a participant belong to one outer fold.
λ*; OOF	Inner-CV selected penalty; out-of-fold prediction	Refit on T_o using λ*; predict V_o once. Full-data ComBat remains upstream.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Nested validation: what is fitted where

1 Hold out outer participants

$T_o \cap V_o = \varnothing$ at participant level

Five outer folds; V_o is never used for scaling or tuning

2 Tune inside T_o only

$\lambda^* = \text{argmin}_\lambda \text{ mean}_{\text{inner}} \text{ MAE}(\lambda)$
(λ^*, w^*) for weighted fusion

Refit scaler inside each of four grouped inner splits

3 Refit on all of T_o

$\mu_o, \sigma_o \leftarrow X[T_o]$
$f_o \leftarrow \text{Ridge}(\lambda^*; T_o)$

Apply frozen training statistics and coefficients to V_o

4 Pool held-out predictions

$\hat{y}_i, \text{OOF} = f_o(x_i),\ i \in V_o$
$\text{MAE} = \text{mean}_i\ |\hat{y}_i, \text{OOF} - y_i|$

Each scan contributes one outer-test prediction; visits share a fold

o: outer fold; T_o: outer-training scans; V_o: outer-test scans; inner: participant-grouped tuning partitions.

This nesting applies to prediction models; upstream full-data harmonization is not nested.

Ridge_GNN_BAG.py; Ridge_fMRI_BAG.py; fusion scripts

Single-modality GNN pipeline

1 Construct node features

$h_v^{(0)} = A[v, :] \in \mathbb{R}^{84}$

DWI nonnegative; fMRI signed profiles

2 Select positive topology

$E_r = \{(u,v): A_{uv} > 0, A_{uv} \geq Q_{1-r}(A^+)\}$

Compare top 10 / 25 / 50 / 75 / 100%

3 Aggregate messages

$m_v^{(l)} = \Sigma_{u \in N(v)} \mathrm{ReLU}(h_u^{(l)} + W_l e_{uv})$

Project scalar edge weights into feature space

4 Update ROI embeddings

$h_v^{(l+1)} = \mathrm{MLP}_l((1 + \varepsilon_l)h_v^{(l)} + m_v^{(l)})$

Three GINE layers + normalization / activation

5 Pool and predict

$g = \mathrm{mean}_v\, h_v^{(3)}$

$\hat{y} = \mu_y + \sigma_y \cdot \mathrm{Head}(g)$

CN grouped OOF; select sparsity by MAE

NOTATION AT A GLANCE

v,u; $N(v)$; E_r ROI nodes; neighbors of v; selected edge set

$h_v^{(0)}$; $h_v^{(l)}$ 84-value connectivity row; layer-l embedding

e_{uv}; W_l Scalar edge weight; learned edge projection

Selected topology: DWI top 50%; fMRI top 10%. Selection and reporting use the same OOF comparison.

GNN_DWI_BAG.py; GNN_fMRI_BAG.py

Single-modality GNN: notation

Symbol / object	Definition and dimension	How to interpret it
v,u; $N(v)$; E_r	ROI nodes; neighbors of v; selected edge set	Use r for retained fraction to distinguish it from a participant index.
$h_v^{(0)}$; $h_v^{(l)}$	84-value connectivity row; layer-l embedding	$h^{(0)} \in \mathbb{R}^{84}$; hidden vectors $\in \mathbb{R}^{64}$; three GINE layers.
e_{uv}; W_l	Scalar edge weight; learned edge projection	Project scalar edges to the current node-feature dimension before addition.
$m_v^{(l)}$; ε_l	Summed neighbor message; self-feature coefficient	ε is fixed at PyG default zero; the MLP transforms self + neighbor information.
g; Head(g)	64-value mean-pooled graph embedding; scalar output	Regressor: 64 → 32 → 1. Its output is standardized age.
μ_y; σ_y; $\hat{y}$	Training age mean / SD; predicted age in years	MSE trains standardized age; grouped inner validation selects epochs.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

GINE: from connectivity to predicted age

1 Form edge-aware messages

$m_v^{(l)} = \Sigma_{u \in N(v)} \mathrm{ReLU}(h_u^{(l)} + W_l e_{uv})$

Layer 0: $h \in \mathbb{R}^{84}$; later layers: $h \in \mathbb{R}^{64}$; edge e is scalar

2 Mix self and neighbor information

$a_v^{(l+1)} = \mathrm{MLP}_l((1+\varepsilon_l)h_v^{(l)} + m_v^{(l)})$

ε_l=0 in this configuration; learned W_l matches feature dimension

3 Normalize and activate

$h_v^{(l+1)} = \mathrm{ReLU}(\mathrm{BN}_l(a_v^{(l+1)}))$

Three layers; dropout after the first two; BN = batch normalization

4 Predict standardized age

$g = (1/84)\Sigma_v h_v^{(3)}$

$\hat{y} = \mu_y + \sigma_y \mathrm{Head}(g)$

Head: 64→32→1; train MSE against $(y-\mu_y)/\sigma_y$

v/u: ROI nodes; l: message-passing layer; N(v): selected neighbors; μ_y/σ_y: inner-training age statistics.

Inner 20% grouped validation selects checkpoints; sparsity selection reuses the reported pooled OOF comparison.

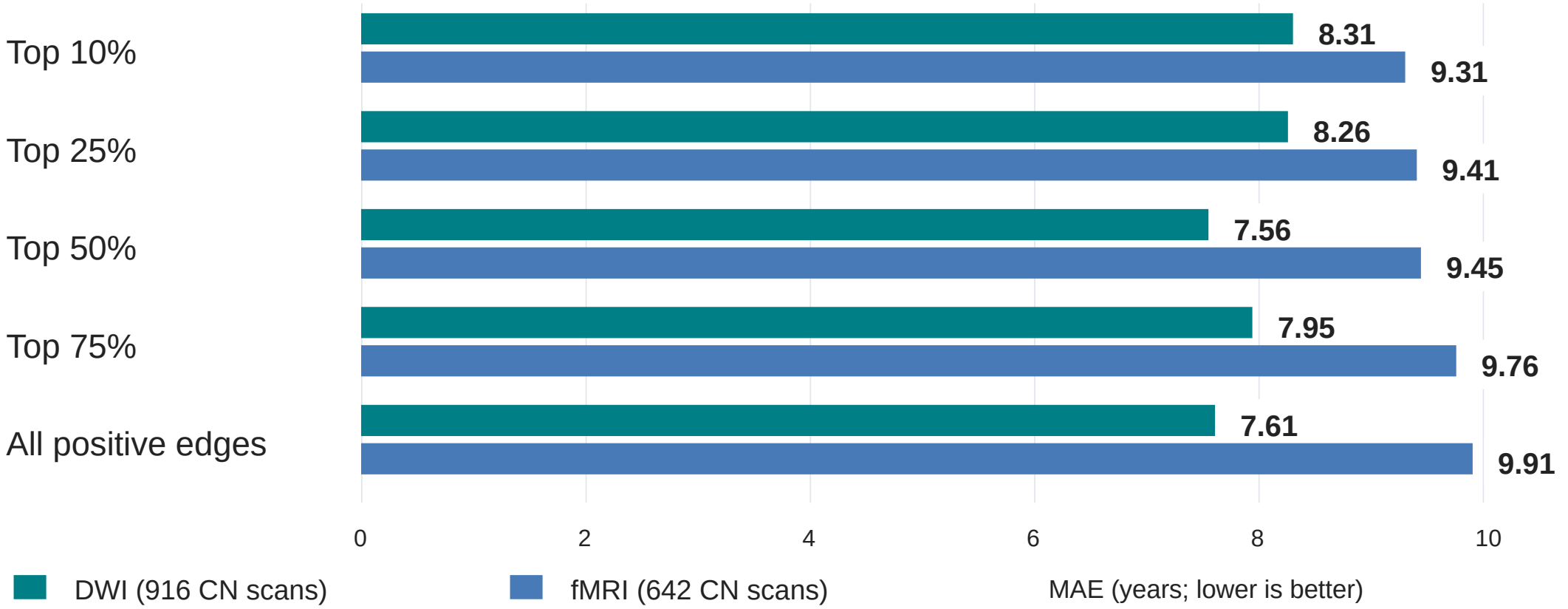
Ridge leads the full-cohort unimodal analyses

Modality / model	CN scans	MAE (yr)	RMSE (yr)	R^2	Pearson r
DWI Ridge	916	5.98	7.89	0.519	0.724
DWI GNN (top 50%)	916	7.56	9.61	0.287	0.537
fMRI Ridge	642	7.33	9.49	0.406	0.637
fMRI GNN (top 10%)	642	9.31	11.97	0.056	0.305

DWI and fMRI use different populations; compare models within each modality.

DWI_Ridge_CV_metrics.json; fMRI_Ridge_CV_metrics.json; selected GNN threshold summaries

Graph sparsity has modality-specific effects



Thresholds were selected on the reported OOF comparison; an independent estimate is still needed.

DWI_GNN_threshold_comparison.csv; fMRI_GNN_threshold_comparison.csv

BAG and age-bias correction pipeline

1 Collect OOF predictions

$\hat{y}_i = f_fold(i)(x_i)$

CN participant excluded from training

2 Compute raw BAG

$BAG_i = \hat{y}_i - y_i$

Positive: predicted older than actual age

3 Estimate age bias

$(\hat{a},\hat{b}) = \text{argmin}_{a,\beta}$
$\Sigma CN\text{-}OOF[BAG_i - (a + by_i)]^2$

Fit age-related trend on the CN reference

4 Subtract age trend

$cBAG_i = BAG_i - (\hat{a} + \hat{b}y_i)$

Age-adjusted descriptive brain age gap

5 Summarize by cohort

mean cBAG = $(1/n)\ \Sigma_i\ cBAG_i$

Separate OOF from final in-sample predictions

NOTATION AT A GLANCE

$f_fold(i)$ Predictor trained without scan i's outer fold

$\hat{y}_i$; y_i Predicted age; chronological age, both in years

BAG_i Raw brain age gap = $\hat{y}_i - y_i$

Independent or cross-fitted calibration is needed for stronger downstream inference.

Ridge CV summaries; GNN threshold summaries

BAG and age-bias correction: notation

Symbol / object	Definition and dimension	How to interpret it
f_fold(i)	Predictor trained without scan i's outer fold	Participant-level grouping also excludes other visits from that participant.
$\hat{y}_i$; y_i	Predicted age; chronological age, both in years	ŷ is the raw prediction before subtracting the BAG age trend.
BAG_i	Raw brain age gap = $\hat{y}_i - y_i$	Positive values indicate older predicted age relative to chronological age.
a; b	Age-bias intercept (years); slope (years/year)	These are calibration coefficients, distinct from Ridge β and intercept.
â; $\hat{b}$; $cBAG_i$	Fitted coefficients; $BAG_i - (\hat{a}+\hat{b}y_i)$	The CN OOF population defines the reference age trend.
Reference vs inference	Descriptive calibration vs independent validation	Fitting and assessing correction on one reference does not independently validate it.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Implementation-verified notation | See preceding diagram and speaker notes.

BAG captures prediction error; age bias needs care

- BAG = predicted brain age − chronological age. Positive BAG means an older predicted age.
- cBAG subtracts the linear age-related BAG trend estimated from CN OOF predictions.
- Paired raw BAG–age correlations remain strongly negative: DWI Ridge −0.599; weighted Ridge −0.616; GNN fusion −0.825.
- A correction fitted and assessed on the same OOF reference is descriptive; downstream inference needs appropriate independent or cross-fitted calibration.
- Full-cohort CN predictions are in-sample after final refitting; use OOF outputs for performance comparisons.

Ridge CV summaries; report_tables/Table4_models_pooled.csv

Cross-modality pairing and training selection

1 Pair modalities

$key^{D}_{i} = key^{f}_{i}$

Dataset + participant + normalized visit

2 Audit ages

$d_i = |age^{D}_{i} - age^{f}_{i}|$

Record conflicts; do not adjudicate silently

3 Apply CN policies

$CN_i = CN^{D}_{i} \wedge CN^{f}_{i}$

Retain the existing DWI HABS override

4 Retain eligible scans

$eligible_i = CN_i \wedge (d_i \leq 0.01) \wedge$ no explicit sex conflict

Age tolerance in years; missing sex allowed

5 Freeze shared folds

n = 387 scans; P = 330

Same participant-grouped outer folds

NOTATION AT A GLANCE

D; F Superscripts for DWI and fMRI

key_i Dataset / participant / normalized visit key

d_i; τ Absolute age disagreement; tolerance $\tau = 0.01$ yr

The same participant-grouped outer folds are reused across the paired model comparisons.

prepare_DWI_fMRI_fusion_metadata.py; metadata/fusion_metadata_audit_summary.json

Cross-modality pairing and training selection: notation

Symbol / object	Definition and dimension	How to interpret it
D; F	Superscripts for DWI and fMRI	Superscripts identify modalities; they are not mathematical powers.
key_i	Dataset / participant / normalized visit key	Key agreement does not establish acquisition-date proximity.
d_i; τ	Absolute age disagreement; tolerance τ = 0.01 yr	Exclude unresolved ages from training; do not silently replace them.
CN^D; CN^f	Modality-specific cognitively normal indicators	DWI HABS all-CN override is retained; fMRI requires explicit CN.
$\wedge$; sex_ok	Logical AND; no explicit sex disagreement	Every required condition must hold; missing sex is allowed in this model.
n; P; q(i)	387 scans; 330 participants; scan-to-participant map	Pairing yields 847 scans; policy and consistency filters define the evaluated subset.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Implementation-verified notation | See preceding diagram and speaker notes.

Paired-sample selection changes the population

Cohort	Paired scans	Evaluated CN scans	Evaluated participants
ADNI	245	131	74
ADRC	338	243	243
HABS	264	13	13
Total	847	387	330

250 HABS scans have age disagreements between modalities. Training requires ages to agree within 0.01 year; only 13 HABS scans enter this benchmark.

Visit-ID pairing is available; acquisition-date proximity has not been verified.

report_tables/Table1_samples.csv; metadata/fusion_metadata_audit_summary.json

Fusion combines modalities in three ways

Approach	Implementation
Ridge early fusion	Concatenate standardized DWI and fMRI edges: 6,972 features
Weighted Ridge early fusion	[Z_DWI, w × Z_fMRI]; tune w and alpha jointly within training folds
Ridge late mean	Average the two unimodal age predictions
GNN fusion	Separate two-layer GINE encoders; concatenate pooled embeddings; nonlinear age head

Fusion GNN uses fixed top 50% positive edges for both modalities; hidden size = 32.

Ridge_baseline/fusion_summary.json; Weighted_Ridge/fusion_summary.json; GNN_fusion/GNN_fusion_summary.json

Ridge fusion pipelines: early, weighted and late

EARLY

Features
x^D, x^f
3,486 edges each

→ **Scale**
$Z = (X - \mu)/\sigma$
Training splits only

→ **Join**
$Z^E = [Z^D ; Z^f]$
6,972 features

→ **Tune penalty**
$\min \Sigma(y - b - Z^E\beta)^2 + \lambda\|\beta\|^2$
Grouped inner CV

→ **Predict**
$\hat{y} = \hat{b} + Z^E\hat{\beta}$
Held-out outer fold

WEIGHTED

Features
x^D, x^f
Same paired CN scans

→ **Scale**
$Z = (X - \mu)/\sigma$
Before block weighting

→ **Weight & join**
$Z^W = [Z^D ; wZ^f]$
w is a feature scale

→ **Tune jointly**
$(\lambda^*, w^*) = \text{argmin}$ MAE_inner(λ, w)
Grouped inner CV

→ **Predict**
$\hat{y} = \hat{b} + Z^W\hat{\beta}$
Refit selected λ and w

LATE MEAN

Features
x^D, x^f
Same outer folds

→ **Separate tuning**
$f^D = \text{Ridge}^D$
$f^f = \text{Ridge}^f$
Scaler / λ per modality

→ **Separate ages**
$\hat{y}^D = f^D(x^D)$
$\hat{y}^f = f^f(x^f)$
Held-out predictions

→ **Average**
$\hat{y} = (\hat{y}^D + \hat{y}^f)/2$
Equal prediction weights

→ **BAG**
BAG = $\hat{y} - y$
Report pooled OOF metrics

D / F: DWI / fMRI; μ / σ: training statistics; λ: Ridge penalty; w: fMRI block weight.

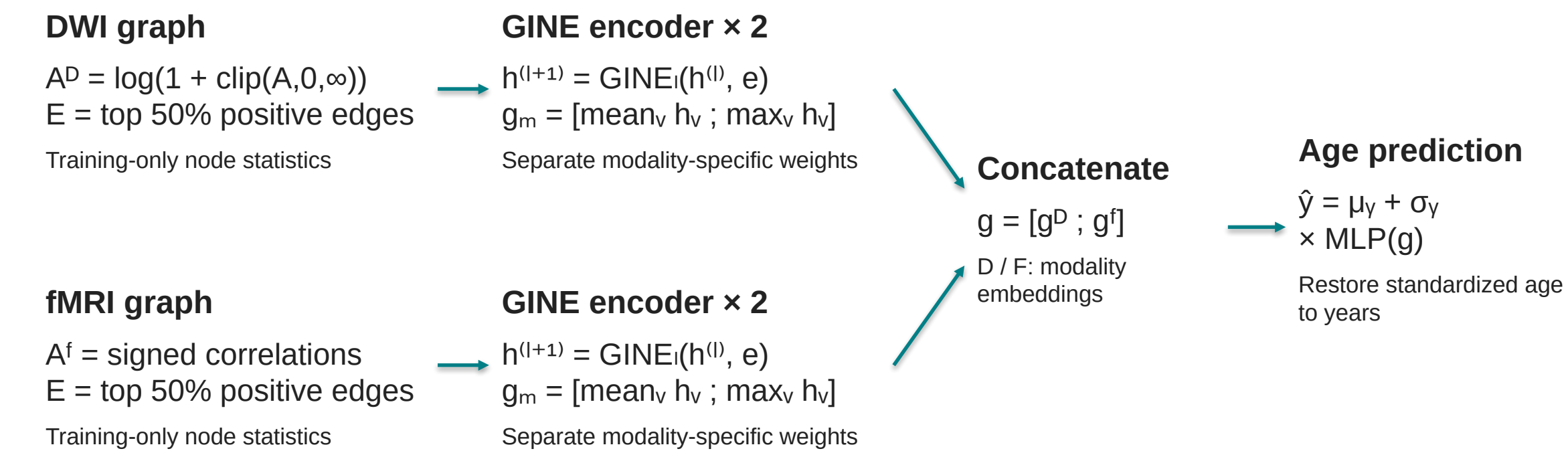
Ridge_DWI_fMRI_fusion.py; Ridge_DWI_fMRI_weighted_fusion.py; fusion_transformers.py

Ridge fusions: early, weighted and late: notation

Symbol / object	Definition and dimension	How to interpret it
Z^D; Z^f	DWI and fMRI standardized edge vectors	Each has 3,486 columns; fit modality columns using training data only.
[;]; Z^E	Feature concatenation; early-fusion vector	$Z^E = [Z^D;Z^f]$ contains 6,972 features.
w; Z^W	fMRI block scale; weighted concatenation	$Z^W=[Z^D;wZ^f]$; DWI scale is one; w is not a contribution percentage.
λ*; w*	Penalty and weight jointly selected by inner CV	Search w ∈ {0,0.25,0.5,1,2,4}; λ is Ridge alpha.
β; b	6,972 edge coefficients; scalar intercept	Only β is penalized; never standardize again after block weighting.
$\hat{y}^D$; $\hat{y}^f$; ŷlate	Two unimodal OOF ages; arithmetic mean	Late fusion averages predictions in years, not coefficients or embeddings.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

GNN fusion pipeline: two modality-specific encoders



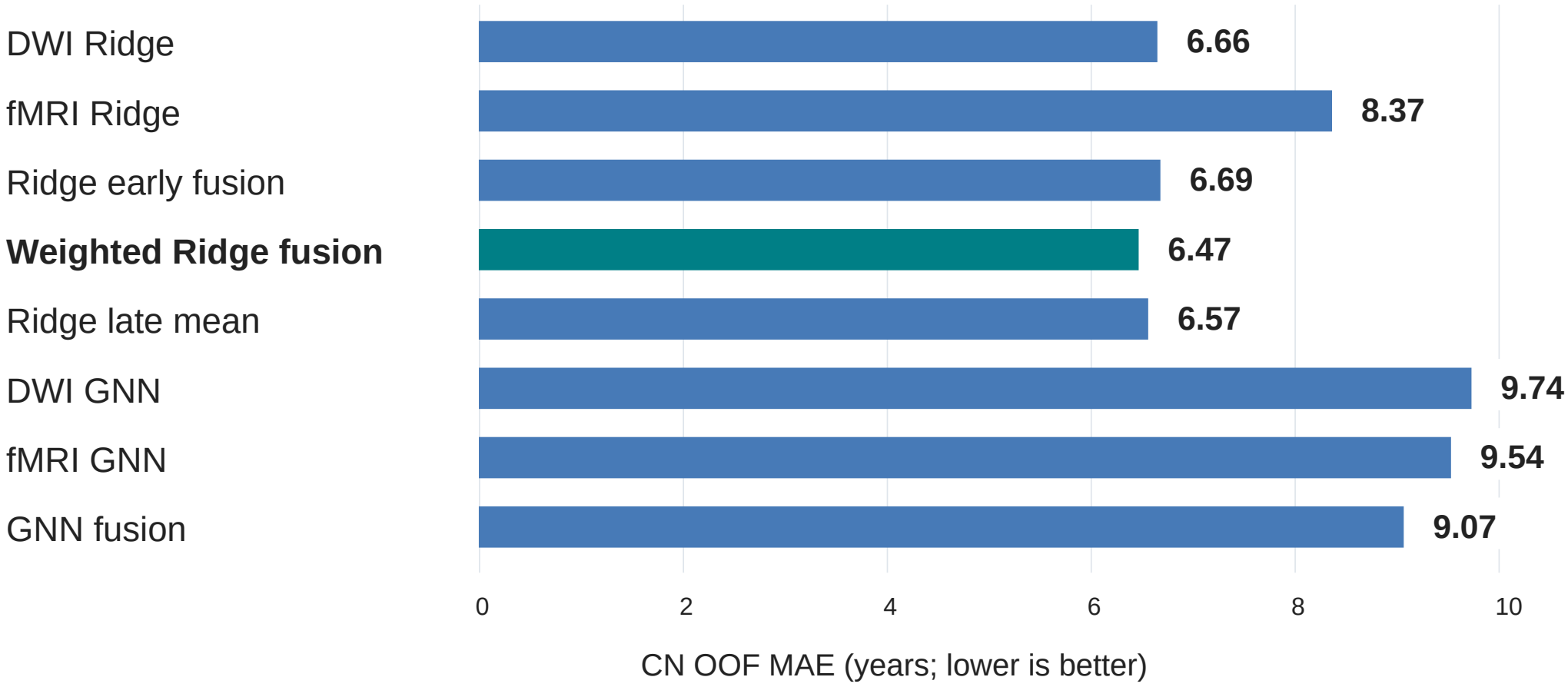
Inner grouped validation selects epochs → fresh outer-training refit → held-out prediction.

GNN fusion: two modality-specific encoders: notation

Symbol / object	Definition and dimension	How to interpret it
$H_m^{(0)}$; m	Node input matrix; modality m ∈ {D,F}	84 × 84 input per modality; DWI log1p weights, fMRI signed correlations.
E_m; $e_{uv,m}$	Positive top-50% topology; normalized scalar weight	e = selected positive weight / graph positive maximum; bidirectional edges.
$H_m^{(l)}$; g_m	ROI embeddings; pooled modality vector	Two GINE layers, hidden 32; mean+max pooling makes $g_m \in \mathbb{R}^{64}$.
$g=[g^D;g^f]$	Concatenated fusion representation ∈ $\mathbb{R}^{128}$	A separate encoder is learned for each modality; weights are not shared.
Head; μ_y; σ_y	128 → 32 → 1 MLP; training age mean and SD	Head predicts standardized age; multiply by σ_y and add μ_y for age years.
$\mu_{m,j}$; $\sigma_{m,j}$	Training node-feature column statistics	Pool training graph nodes per modality; fresh outer refit follows inner epoch selection.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Weighted Ridge has the lowest paired-sample MAE



Shared benchmark: 387 CN scans / 330 participants / identical outer folds.

report_tables/Table4_models_pooled.csv

The fusion gain is modest and method dependent

Model	MAE (yr)	RMSE (yr)	R^2	r
DWI Ridge	6.66	8.49	0.620	0.788
Ridge early fusion	6.69	8.62	0.609	0.780
Weighted Ridge fusion	6.47	8.38	0.630	0.794
Ridge late mean	6.57	8.48	0.621	0.799
GNN fusion	9.07	11.44	0.311	0.558

Weighted Ridge improves MAE by 0.184 years (2.8%) vs DWI Ridge. Unweighted concatenation does not improve on DWI alone.

report_tables/Table4_models_pooled.csv; Weighted_Ridge/fusion_summary.json

Pooled performance masks cohort differences

Cohort	CN scans	DWI Ridge MAE	Weighted Ridge MAE	GNN fusion MAE
ADNI	131	7.15	6.85	8.91
ADRC	243	6.52	6.39	9.04
HABS	13	4.12	4.16	11.24

ADNI R^2 is negative for all paired models despite positive pooled R^2. HABS estimates rely on only 13 scans. These are pooled-training OOF results, not leave-one-cohort-out tests.

report_tables/Table5_models_by_cohort.csv

Interpretability pipeline: from OOF models to attribution

1 Load original fold model

$f = f_{saved,fold}$

Original weights + normalizers; no retraining

2 Verify predictions

$\Delta_i = f(x_i) - \hat{y}original,_i$

Inspect differences before interpretation

3 Define reference path

$x(t) = x' + t(x - x')$

Training-mean node / zero-edge reference

4 Integrate gradients

$IG_k = (x_k - x'_k) \times \int_0^1 \partial f(x(t))/\partial x_k \, dt$

Node + edge attribution; fixed graph topology

5 Check completeness

$r = f(x) - f(x') - \Sigma_k IG_k$

Tolerance: $|r| \leq 0.05$ yr; up to 512 steps

NOTATION AT A GLANCE

f; x; x′ Saved outer-fold age predictor; observed input; reference
x′ node / edge Training-mean node values; zero edge attributes
k; t; x(t) Input coordinate; path fraction; interpolated input

Prediction gate: checkpoint ↔ OOF CSV ≤ 0.0001 yr; inference ↔ checkpoint ≤ 0.001 yr.

Interpretability: from OOF models to attribution: notation

Symbol / object	Definition and dimension	How to interpret it
f; x; x′	Saved outer-fold age predictor; observed input; reference	f outputs age years; x jointly contains both modalities' node and edge attributes.
x′ node / edge	Training-mean node values; zero edge attributes	Node mean corresponds to standardized zero; keep observed diagonal fixed.
k; t; x(t)	Input coordinate; path fraction; interpolated input	k is not a fold; $t \in [0,1]$; $x(t)=x'+t(x-x')$; topology is unchanged.
$\partial f/\partial x_k$; IG_k	Local model sensitivity; integrated contribution	IG has age-year units; it explains $f(x)-f(x')$, not a causal effect.
S; rIG	Integration intervals; completeness residual	Trapezoidal quadrature; S = 64 initially, adapt up to 512; $\lvert rIG \rvert \le 0.05$ yr.
Δpred; rIG	Prediction consistency vs integration error	Require inference/checkpoint difference ≤ 0.001 yr; IG completeness is assessed separately.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Implementation-verified notation | See preceding diagram and speaker notes.

Integrated gradients: reference and numerical checks

1 Define an explicit reference

x'_{node} = training mean; $x'_{edge} = 0$
$x(t) = x' + t(x - x')$

Keep observed node diagonal and edge topology; interpolate both modalities

2 Integrate each input coordinate

$IG_k = (x_k - x'_k) \int_0^1 \partial f(x(t))/\partial x_k \, dt$

Trapezoidal quadrature; f outputs chronological-age prediction in years

3 Check integration and prediction

$rIG = f(x) - f(x') - \Sigma_k IG_k$
$\Delta pred = f(x) - \hat{y}_{original}$

Adapt 64→512 intervals; flag |rIG|>0.05 years; verify prediction consistency separately

4 Map inputs to connections

$A_{ij} = IG_{node,ij} + IG_{node,ji} + IG_{edge,i\to j} + IG_{edge,j\to i}$

Then average signed visits per participant before absolute importance

k: transformed-input coordinate; t: path position; i/j: ROI endpoints; f: saved outer-fold predictor.

The current code explains original saved OOF models; corresponding verified outputs are not yet available here.

GNN_BAG_interpretability.py; analysis_summary.json

Current attribution code and available outputs

Current code: load original outer-fold checkpoints → verify predictions → compute IG.

Each checkpoint contains the learned weights, node normalizers, age statistics, fold/sample identity and held-out predictions.
The current workspace does not yet contain the required outer_fold_models directory.
The following attribution figures are retained as archived reconstructed-model results; their 13.18-year mismatch is not a property of the updated checkpoint workflow.

No new attribution results or improved consistency were claimed without running and verifying the current workflow.

Archived attribution: reconstructed fusion GNN

- Integrated gradients (IG) attributes standardized node features and normalized edge attributes on held-out CN scans.
- Five reconstructed outer-fold models explain 387 scans from 330 participants.
- Reference: training-mean node features with observed diagonal retained, and zero edge attributes; graph topology stays fixed.
- Participant summaries average signed attributions across visits before taking absolute values.
- Attributions describe model behavior relative to this reference; they are not causal effects or clinical associations.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/METHODS.txt; analysis_summary.json

Interpretability summaries and robustness checks

1 Average visits

$\bar{A}_{p,ij} = \text{mean}_s \text{visits}(p)\ A_{s,ij}$

Signed connection IG; equal participant weight

2 Rank connections

$I_{ij} = \text{mean}_p\ |\bar{A}_{p,ij}|$

Take absolute value after visit averaging

3 Score regions

$ROI_i = \frac{1}{2} \sum_{j \neq i} I_{ij}$

Allocate half to each connection endpoint

4 Check rank stability

$S_{ij} = \text{countfolds}(\text{rank}_{ij} \leq 100)$

ROI: top 10; connections: top 100

5 Reference occlusion

$\Delta MAE_m = \text{mean}_p\ \text{mean}_s$
$[|f(x_{-m}, x'_m) - y| - |f(x) - y|]$

Replace one modality; positive = higher error

NOTATION AT A GLANCE

$A_{s,ij}$; p; V_p Scan connection IG; participant; participant visits

A node; A edge Node-profile attribution; edge-attribute attribution

$\bar{A}_{p,ij}$; I_{ij} Signed visit average; mean participant absolute score

Model reliance and rank stability are descriptive; neither establishes causal or clinical relevance.

results/explore/GNN_attribution/*.csv; analysis_summary.json

Interpretability summaries and robustness checks: notation

Symbol / object	Definition and dimension	How to interpret it
$A_{s,ij}$; p; V_p	Scan connection IG; participant; participant visits	i,j now index ROI endpoints; s indexes scans to avoid an index collision.
A node; A edge	Node-profile attribution; edge-attribute attribution	Sum both symmetric node entries and both directed edge entries for each pair.
$\bar{A}_{p,ij}$; I_{ij}	Signed visit average; mean participant absolute score	Average signed scan values first, then take absolute values across participants.
ROI_i	$\frac{1}{2}\,\Sigma_{j\neq i}\, I_{ij}$, in attribution years	Each connection contributes half its absolute importance to each endpoint.
rank; top-k frequency	Per-fold ranking; count of folds in top k	k = 100 for connections; k = 10 for ROIs; frequency ranges from 0 to 5.
x_{-m},x'_m; ΔMAE_m	Keep other input; replace modality m; error change	Average within participant, then across participants; positive = worse prediction.

Indices and units are local to this diagram; modality superscripts are labels, and hats indicate fitted estimates.

Attribution quality limits the strength of interpretation

Quality check	Observed result
IG integration steps	64 initially; up to 512
Completeness tolerance	0.05 years
Scans failing completeness	6 / 387 (1.6%)
Maximum absolute IG residual	0.119 years
Maximum difference from original OOF prediction	13.18 years

These attributions explain reconstructed models; exact original-model attribution is not established.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/analysis_summary.json; reconstruction_check.csv

fMRI carries larger attribution in this fusion model

Modality	Mean signed IG (yr)	Mean absolute connection IG	Occlusion error increase (yr)
DWI	-6.96	45.17	2.55
fMRI	10.33	128.54	3.36

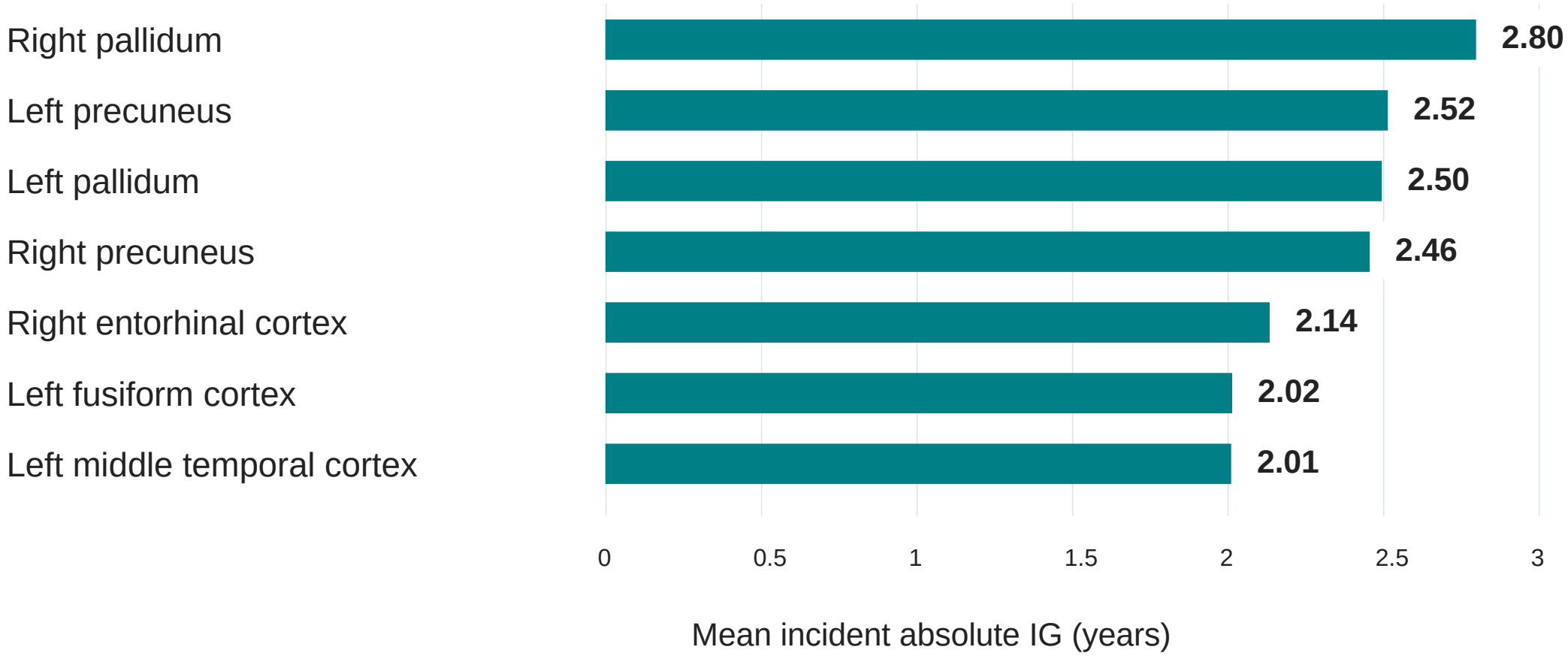
Larger fMRI attribution reflects this model and its reference. It does not imply superior standalone prediction. Absolute connection contributions can accumulate and cancel; they are not predicted ages or modality percentages.

Reference occlusion retains topology and may produce inputs outside the training distribution.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/modality_summary.csv

Top ROI attributions emphasize fMRI regions



Top overall rows are fMRI. Region labels follow the existing atlas audit; per-scan order is unverified.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/roi_importance.csv; roi_labels_with_verification_scope.csv

Leading connections and stability offer complementary views

Connection (fMRI)	Mean |IG| (yr)	Folds in top 100
lh_precuneus ↔ rh_precuneus	0.415	5/5
lh_entorhinal ↔ rh_entorhinal	0.233	5/5
lh_isthmuscingulate ↔ lh_precuneus	0.195	5/5
lh_inferiorparietal ↔ lh_precuneus	0.183	3/5
lh_insula ↔ rh_insula	0.170	5/5

Ranks describe reconstructed CN models and fixed topology; avoid biomarker claims.

Historical results: results/explore/GNN_attribution. Not the output of the current saved-checkpoint workflow.

GNN_interpretability/connection_importance.csv; connection_fold_stability.csv

What the current evidence supports

- Technical dataset/site information is substantially reduced after harmonization, with mixed preservation diagnostics.
- Ridge is the stronger baseline in these experiments; weighted fusion yields a small paired-sample improvement.
- GNN fusion improves on the paired unimodal GNNs, but remains behind Ridge.
- Archived attribution suggests exploratory candidates; verified original-checkpoint attribution results remain pending.
- Next validation priorities: fold-contained harmonization, independent cohort testing, repeated seeds, and exact saved-checkpoint attribution.

Result provenance and run selection

Section	Primary saved evidence
Harmonization	final/Fusion/report_tables/Table3_harmonization.csv
DWI single modality	final/DWI/{Ridge_BAG,GNN_BAG}/*summary*.json / *metrics.json
fMRI single modality	final/fMRI/{Ridge_BAG,GNN_BAG}
Shared fusion benchmark	final/Fusion/report_tables/Table1,4,5,6*.csv
Interpretability	explore/GNN_attribution/*.csv; analysis_summary.json

Current selected-threshold summaries are used for unimodal GNNs. Older 429-scan DWI summaries and legacy fMRI Ridge outputs are kept separate.

Paths are relative to the project results directory.